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Metro tunnelling in Singapore – Thomson Line tunnelling in close proximity to existing railway lines

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ABSTRACT: Singapore's 30km-long all-underground Thomson Line (TSL) involves the operation of approximately thirty Tunnel Boring Machines (TBMs) to complete the twin tunnels and the construction of twenty-two underground stations, including five interchange stations. This involves challenges such as tunnelling in urbanized areas and poor geological conditions such as reclaimed land. Bored tunnels of 5.8m internal diameter are aligned in side-by-side or stacked configuration to accommodate these constraints. In addition to challenges posed by densely-urbanised landscape, the project runs through a number of interfaces, which necessitates tunnelling in close proximity to existing railway lines and stations, such as interface at Outram Park and Robinson Road. Another challenging construction contracts, T226, includes the construction of a portion of Marina Bay Station, a TBM launch shaft, two working shafts to facilitate the underpinning of live operational rail tunnels and station, mining of a pedestrian link directly underneath with clearance of 1.0m, as well as mined mainline tunnels.

1 PROJECT INTRODUCTION AND GENERAL OVERVIEW OF THOMSON LINE

The Thomson East Coast Line (TEL) is the sixth Mass Rapid Transit (MRT) line in Singapore and will be fully automated and driverless. The line will be opened in 2019 onwards over five stages. It is expected to eventually service one million commuters daily when fully operational. The entire TEL stretches over 43km and the entirety of construction will be subsurface, with thirty-one stations and seven interchange stations. The alignment runs from the north to south, cutting through the Singapore's CBD and to the east. The alignment will connect with the existing MRT lines at several locations.

The Thomson Line (TSL) is a portion of the TEL line which runs from the north to the south of Singapore between Woodlands North and Gardens by the Bay. The project owner and developer, Singapore's Land Transport Authority (LTA) has divided the TSL line into four contract packages for engineering consultancy services. Arup was successfully appointed to two out of the four packages, TSL Package A (TSL-A) at the north, and TSL Package D (TSL-D) at the south. TSL-A begins at Woodlands North Station and consists of three stations and associated tunnels. TSL-D is located at the southern tip of the TSL, and interfaces with the Eastern Regional Line (ERL) at Garden by the Bay Marina East shaft along the east coast of Singapore to form the overall TEL. This paper focuses on TSL-D. Figure 1 below illustrates the overall alignment of TEL in relation to the existing MRT lines in Singapore.

TSL-D consists of approximately two nos. of 6km running tunnels and six stations, beginning at the southern end of Havelock Station and ending at the intervention shaft near Gardens by the Bay Marina East shaft for future connection with the ERL. To enhance

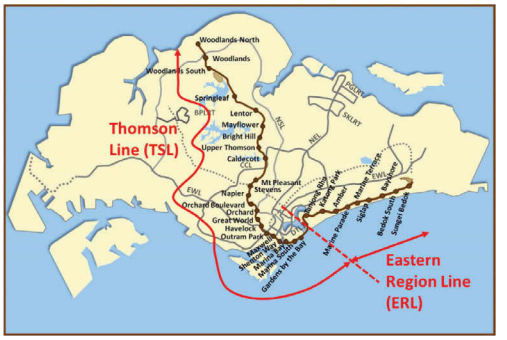


Figure 1. Overall TEL Alignment in relation to Existing MRT Lines.

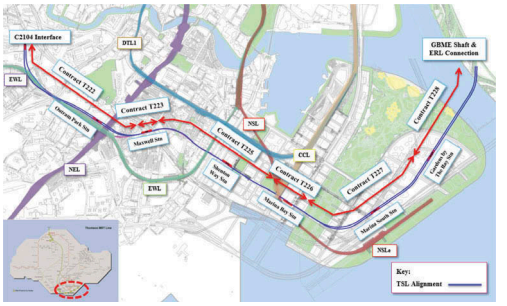


Figure 2. Overall Alignment of TSL-D and Construction Contract Packages.

connectivity, TSL-D will interface with four existing MRT lines at two existing stations namely Outram Park Station and Marina Bay Station. Since the alignment of TSL-D runs through the CBD, the twin bored tunnels of 5.8m internal diameter are aligned in stacked configuration at some locations to accommodate the constraints of the densely-built urban environment. Figure 2 below illustrates the overall alignment of TSL-D, including the different contract packages for station and tunnels construction.

2 GEOLOGICAL CONDITIONS OF SINGAPORE

Singapore's geological formations can be broadly categorized into four major classifications: Bukit Timah Granite from the Mesozoic era is commonly encountered in the central region of Singapore; Jurong Formation, which is a sedimentary rock formation from the Mesozoic era,

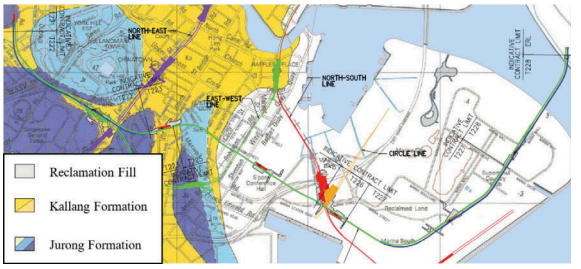


Figure 3. TSL-D Alignment Overlying Singapore's Geological Map.

and is distributed in the western regions of Singapore; Old Alluvium (OA) from the Pleistocene age, and commonly encountered in the east; and the Kallang Formation, which consists of interbedded marine and terrestrial sediment layers, deposited in the Holocene period, most commonly found in the coastal areas and low-lying regions of Singapore.

TSL-D encounters Jurong Formation, Kallang Formation, Old Alluvium, and Fort Canning Boulder Bed (FCBB), which is a colluvial deposit of Jurong Formation material from seismic activity in the past. Towards the southern portion of the TSL-D alignment, construction works encounter reclaimed land, which comprises of consolidating marine clay, fluvial sand and clay of Kallang Formation. Figure 3 below illustrates TSL-D alignment overlying an extract of geological map of Singapore.

3 TUNNELLING NEAR EXISTING RAILWAY LINES AT OUTRAM PARK

One of the main engineering challenges in TSL-D is the running of TSL tunnels close to existing MRT tunnels. In addition, these existing railway lines need to remain in operation throughout the TSL constructions works, which renders the project even more challenging. There were three main instances where the close proximity of tunnels required further study and detailed impact assessment, close monitoring during the project and implementation of additional mitigation measures specified in construction contract, as elaborated in sections below.

3.1 *West of Outram Park (Undercrossing EWL)*

One of the instances is at the western side of Outram Park Station where the TSL tunnel alignment is running underneath the existing EWL tunnels for approximately 410m long stretch. The bored tunnels are in the sedimentary Jurong Formation and have a minimum clearance of 6m to the existing tunnels. Figure 4 below illustrates the location of the undercrossing and minimum clearance between the MRT lines.

Undercrossing of existing tunnels with such minimal clearance could result in settlement and, consequently, potential opening of circumferential joint and water ingress into the tunnel due to gasket relaxation. Ultimately, excessive settlement could impair the railway operation arising from failure of rail fasteners, track sag and twist. Therefore, the impact of TSL tunnelling on the existing EWL needs to be checked against the criteria set-out in the LTA DBC Code of Practice for Railway Protection (CoPRP). The CoPRP allows either compliance with the deemed-to-satisfy provisions or alternative solution achieving performance-based requirements. Arup in-house software Xdisp has been used to calculate the ground movement due to TSL construction works, including TSL bored tunnels and a mined cross passage. The analysis in Xdisp is based on the 3-dimensional Gaussian settlement trough

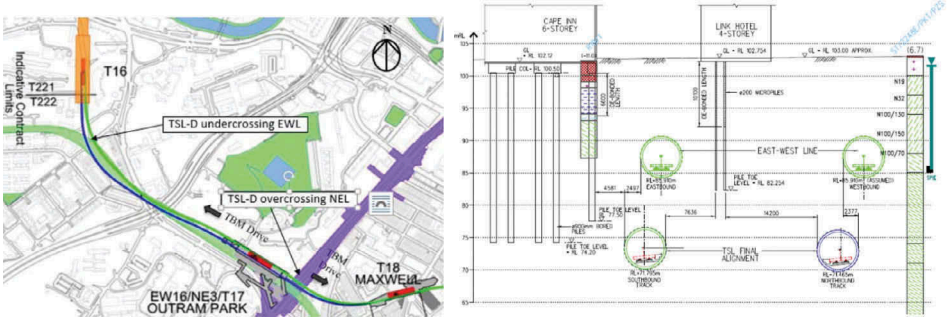


Figure 4. TSL-D, EWL and North East Line (NEL) Interfacing Near Outram Park Station Layout and Section.

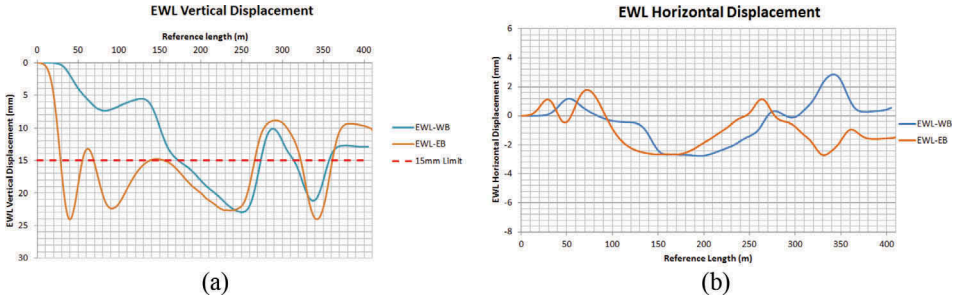


Figure 5. (a) Estimated maximum vertical displacement and (b) maximum horizontal movement of EWL tunnels along interface section.

Table 1. Volume loss values (in %) for 10 limit state assessment.

Model	Full Face Jurong Formation Rock (Grade III or better)	Mixed Face of Jurong Formation Rock/Soil	SCL Construction of Cross Passage 1
Case 1	0.5	1.0	0.5
Case 2	0.5	1.5	0.5
Case 3	1.0	1.5	1.0
Case 4	Mixed face conditions along both TSL bored tunnels and mined CP1 construction, 1.5%		

suggested by Mair et al. (1993). Figure 5 shows the cumulative vertical and horizontal movement of the EWL tunnels estimated using Xdisp. In this analysis the ground movement is taken at the EWL tunnel invert without considering the EWL tunnel structural stiffness, yielding to conservative results.

Nevertheless, the estimated movement of the EWL tunnels exceeds the limit set out by the deemed-to-satisfy provision for EWL tunnels which is 15mm movement in any direction. Hence, an alternative solution involving detailed assessment of the TSL construction impact according to the 10 limit states proposed by Shirlaw et al. (2000) was undertaken. The 10 limit states govern the serviceability and operability of railway tunnels, namely tunnel ovalisation, squat, step, opening of circle joints, additional water make, structural gauge clearance, failure of rail fasteners, track dip/peak, track horizontal versine and track twist. The results of the detailed assessment show that the impacts on the EWL are well within the critical levels for all the 10 limit states. Table 1 shows the volume loss values adopted in the above-mentioned assessment of the EWL and for all cases, the assessment is satisfactory. Case 2 was set as the target performance for construction.

3.2 *East of Outram Park (Parallel to EWL)*

At the east side of the Outram Park station, the TSL runs in parallel to the existing EWL bored tunnels with closest separation of 6m. The related construction works include a launch shaft, a temporary shaft, TSL twin bored tunneling and a mined pedestrian link, as shown in Figure 6. The movement of existing EWL tunnels were estimated using the software Xdisp. Figure 7 (a) shows the cumulative displacements along EWL-EB invert due to various relevant construction works as described above calculated using Xdisp. As shown in Figure 7 (a), the predicted cumulative movement, which is approximately 58mm, exceeds the the limit set out by the deemed-to-satisfy provision for EWL tunnels. Detailed assessment of the construction impact according to the 10 limit states were hence carried out. The impacts on the EWL tunnels are shown to be well within the critical levels for the 10 limit states. Actual settlement at

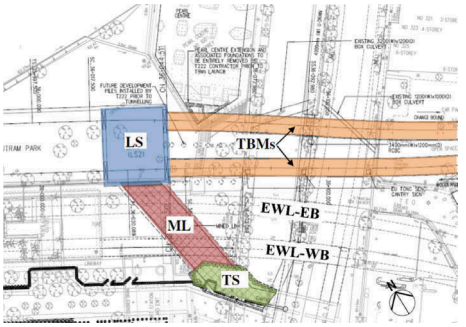
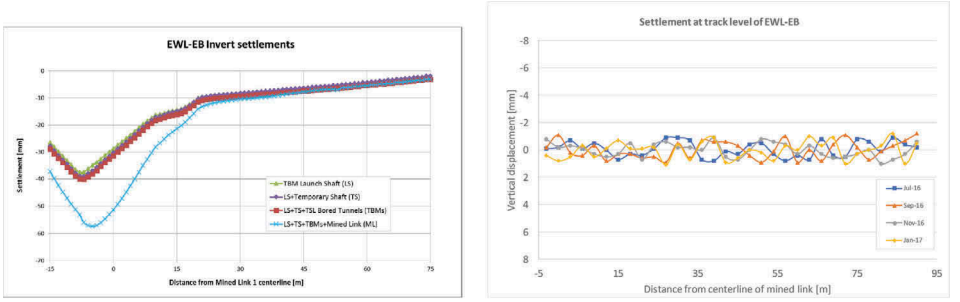


Figure 6. Location of assessed EWL tunnels and related TSL works.



(a) (b)

Figure 7. (a) Cumulative vertical displacement along EWL-EB Invert and (b) Monitoring results of the settlement at track level of the EWL-EB tunnel along its alignment across the construction period.

the invert of the EWL tunnels estimated from the settlement data recorded at the track level were much smaller than predicted, which maximum is shown to be within 2.0mm (Figure 7 (b)).

3.3 *East of Outram Park (Overcrossing NEL)*

The TSL tunnels interface with the NEL line to the east of Outram Park Station, as shown in Figure 4. The TSL-D line over-crosses the NEL tunnels with a minimum clearance of 2.4m in Jurong Formation. Finite element simulation was conducted to predict the movement of the NEL caused by overcrossing of the TSL tunnels. The simulation results show that the NEL tunnel movements and track distortions are within the CoPRP Deemed-to-Satisfy provisions and hence, no further assessment is required. Similarly, NEL tunnels were monitored by ATMS (Automated Track Monitoring System). Monitoring data shows that the movement of the NEL tunnels is below 2mm, which is within the construction alert level.

4 TUNNELLING UNDER-CROSSING EXISTING RAILWAY LINES AT ROBINSON ROAD

The third notable interface is the undercrossing of the EWL tunnels at Robinson Road with a minimum clearance of 3.7m in a mixed face condition of Fort Canning Boulder Bed (FCBB) and Kallang Formation (KF). At this location TSL soil investigation reveals abrupt transition of geological formations from Fort Canning Boulder Bed (FCBB) at the North-West, to Kallang Formation overlying Old Alluvium, at the intersection of existing EWL tunnels and proposed TSL tunnels at the junction of McCallum Street and Robinson Road, as shown in

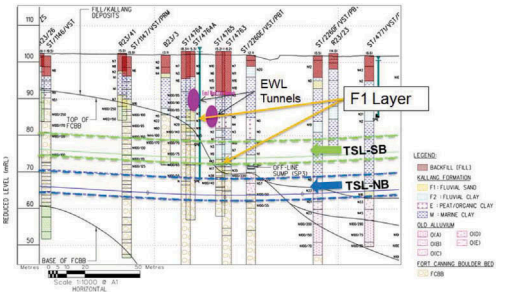


Figure 8. Longitudinal Geological Profile along TSL undercrossing EWL at Robinson road.

Figure 8. Tunneling through drastic geological change, i.e. mixed-face condition poses risk of high ground loss as it is relatively more difficult to balance the penetration rate against the excavation across the tunnel face. The excessive ground loss may potentially jeopardise the stability of the existing EWL tunnels. At this location, the twin TSL tunnels are in stacked configuration.

To mitigate the engineering risks and hazards associated with the close proximity of the tunnels, a variety of measures such as pre-tunnelling protection measures involving ground improvement and micro-tunnelling of canopy pipe piles and TBM control measures at the upper TSL tunnels were proposed.

Fluvial sand layers were also detected along the KF-FCBB interface. Fluvial sand layers act as an aquifer channel which may cause high water ingress into the TBM and groundwater drawdown and consequently, ground settlement at a distance away from the TBM location. In addition, Fluvial sand layers are not easy to treat and treatment may not be effective. TAM grouting was proposed to lower the permeability of the Fluvial sand layers around the TSL upper tunnel and hence, acting as water cut-off.

4.1 *Protective hood of canopy pipe piles*

Arup proposed two temporary shafts, North West (NW) and South East (SE) shafts, to be constructed to facilitate installation of canopy pipe piles (see Figure 9). Steel pipe piles are then jacked from one shaft on one side of existing EWL, steered using slurry Micro TBM underneath the EWL bored tunnels towards the shaft on other side of EWL to retrieve the Micro TBM. The canopy pipe piles are interlocked to each other and fix-ended to the shaft walls. In this manner, the canopy pipe piles serves as a suspended roof which is able to minimise potential localised settlement of the EWL tunnels in the event that large ground loss is induced during TSL upper South Bound (SB) TBM drive.

However, in view of the challenges associated with to divert the massive network of utilities underneath Robinson Road in order to construct the retrieval shaft, another pipe-jacking scheme has been adopted by the contractor. Instead of two shafts, only the SE launch shaft is constructed and the canopy pipe piles are socketed into the FCBB which practically creates fix-end condition similar to the function provided by shaft wall as the Micro TBM proposed has been designed with a retractable cutterhead which omits the need of a 2nd NW shaft designed for retrieval of the Micro TBMs.

4.2 *EPB TBM equipped with AFC*

Experience with EPB TBM in mixed face condition in Singapore indicates potential risk of loss of face pressure and risk of potential sinkhole formation due to inability to anticipate exact location of abrupt geological change at the face and potential sudden drop of face pressure. To mitigate this risk, the EPB TBM is specified with built-in AFC (Automatic Active Face Support System), which will automatically regulate the pressure inside the

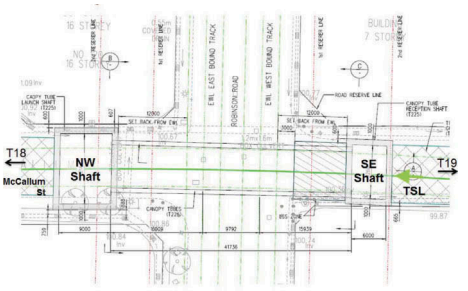


Figure 9. Location of the proposed temporary NW and SE shafts.

plenum in real-time to minimise potential ground loss and hence, movement of the existing EWL tunnels.

4.3 *Settlement analysis considering various construction works and assessment of EWL*

Movements of the EWL tunnels have been calculated considering the effects of various TSL construction activities and mitigation measures. These include NW and SE shafts construction, ground improvement works, canopy pipe piles micro-tunnelling, construction of TSL-NB (lower) and TSL-SB (upper) bored tunnels as well as consolidation settlement.

Canopy pipe piles are prescribed to reduce potential settlement of the existing EWL tunnels in case of ground loss when TSL TBM encounter mixed soil condition underneath the existing EWL tunnels. When the face pressure is at 80% of overburden with the presence of canopy pipe piles, maximum settlement is predicted as 55mm. And this has been assessed against the 10 limit states.

Figure 10 shows the cumulative settlement of the EWL-WB tunnel induced by various construction works with mitigation measures considered. As shown, the movements of both EWL tunnels (maximum 55mm) exceed the CoPRP Deemed-to-Satisfy provisions. Thus, alternative approach was undertaken by detailed assessment of the effects against the 10 Limit States. The results of the alternative assessment shows that the impacts on the EWL are within the critical levels for all the 10 limit states when mitigation measures are adopted.

Figure 11 shows the ATMS monitoring readings of the total movement (vector sum of horizontal and vertical displacement) of the EWL-WB tunnels at different construction stages. Monitoring prisms "225WC19" and "225WR19" are located at the interface with TSL bored tunnels and are hence expected to produce the highest displacement readings (see Figure 11). Figure 12 plots the total movement recorded by different monitoring prisms installed along the EWL-WB alignment. The curves of the monitoring results generally exhibit the shape of a Gaussian settlement trough (Figure 12), suggesting the validity of the data.

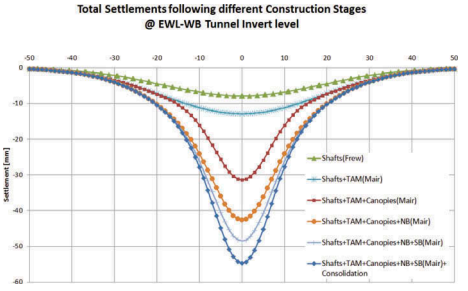


Figure 10. Predicted settlements at EWL-WB invert.

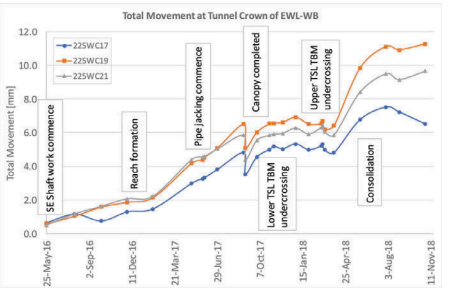


Figure 11. Total movement readings of EWL-WB tunnel at crown across construction period.

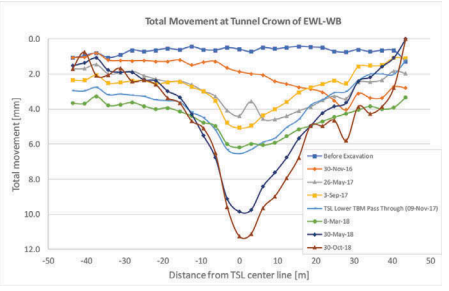


Figure 12. Total movement readings of EWL-WB tunnel at crown along the alignment.

The ATMS-recorded data indicates that the maximum movement of the EWL-WB Tunnel is within 12mm (as shown in Figure 12), which is well below the predicted value (Figure 10). The smaller movement of the EWL tunnel can be attributed to the deviation in protective measures, i.e. constructing only one shaft, lower actual volume loss induced during canopy pipe pile construction. In addition, the estimated ground movement by Xdisp is conservative in nature and was expected to be higher than the actual measurements. The conservativeness of the analysis comes from the fact that the model does not consider the stiffness of the EWL tunnels, mobilised ground arching or soil-structure interaction, which hold significant impact. Nevertheless, the monitoring results indicate the successful implementation of the notional design of the TSL undercrossing EWL.

5 DESIGN AND CONSTRUCTION OF MINED TUNNELS BELOW AN OPERATIONAL THREE-WAY INTERCHANGE STATION AT MARINA BAY

One of the most challenging construction contracts of the TSL-D is contract T226, which included the construction of the TSL Marina Bay Station, a TBM launch shaft, two working shafts to facilitate the underpinning of North South Line (NSL) tunnels, mining of a pedestrian linkway directly underneath the NSL tunnels and station platforms, as well as mined mainline tunnels in a stacked arrangement underneath the pedestrian linkway. It was also stipulated that the design and construction works are to be strictly compliant with the allowable movement of existing tunnels with maximum movement of 15mm, and that no existing MRT services are permitted to be closed or degraded in any way during the construction of works. The consolidating Marine Clay of the Kallang Formation was expected to extend to a depth of 35m. Figure 13 (a) below illustrates the layout of Marina Bay Station and the proposed construction works under Contract T226.

The pedestrian link is 40m in length, 10.7m wide, and 4.1m in height. It was constructed directly underneath the base slabs of the NSL tunnels. The first challenge is to carry out mining safely in the very soft Marine Clay (typical SPT value ranges from 0 to 2). Vertical jet

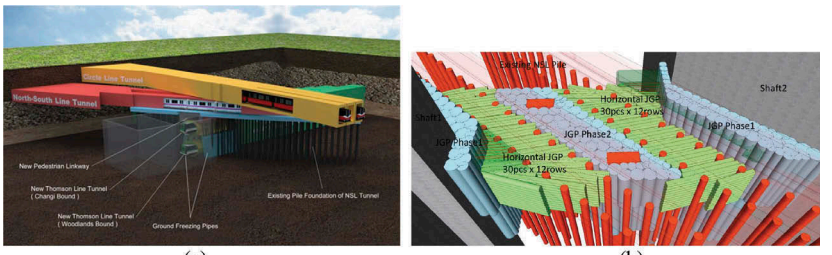
grouting from a limited work site space directly between the NSL and CCL tunnels as well as horizontal jet grouting from working shafts were utilized to improve the ground condition for mining construction. The required minimum undrained shear strength ("S_u" or "C_u") of the treated soil is 300 kPa, as stipulated in Arup's notional design. Figure 13 (b) shows the 3D model of the ground improvement layout. The pedestrian linkway is constructed under a few stages namely:

1. Mining of the two adits using a rectangular open-face shield with compressed air support while cutting NSL piles along the alignment and supporting the adits using temporary steel segments;
2. Mined enlargement of the adits and construction of the transfer beam to underpin the NSL tunnels; and
3. Mining the ground in between the two adits, cut the remaining NSL piles within the linkway space and construct the linkway permanent structure.

Another challenge is the removal of the existing foundation piles of the NSL tunnels during shield mining of the pedestrian linkway. Stitch coring method was adopted to cut the piles as cutting work needs to be carried out in a confined space in the front of the shield machine with limited impact to the NSL tunnels above (Figure 14). A total of 18 piles were cut to construct the pedestrian link. Ground freezing is proposed as groundwater cut-off during mining works for the new TSL tunnels as ground improvement is known to be ineffective in Old Alluvium formation. Transfer beams parallel to the pedestrian linkway have been designed to support the loads of the NSL tunnels since part of the existing pile foundations need to be removed. Flat jacks were provided as a provision to jack up the NSL tunnels in the event of settlement. 4 nos. of flat jacks were provided below the NSL tunnel base slab, as shown in Figure 15.

Real-time monitoring of the settlement of the NSL tunnel soffit was conducted during the shield driving to excavate the adits for the pedestrian linkway, which involves pile cutting work. Monitoring data shows that the actual settlement of the NSL tunnels is within 1 mm for both Adit 1 and Adit 2. This is likely due to the high JGP strength installed below the NSL tunnels which supports the tunnels and prevents excessive settlement. The tests on core samples from the treated soil shows that the achieved undrained shear strength was between 600 kPa to 1000 kPa, which exceeds the minimum design requirement by a large extent. The existing foundation piles were safely removed without adverse effect on the existing NSL tunnels. These indicate the effectiveness of the solutions offered to the project challenges and highlights the success of the concept design.

Minimal displacement of the NSL tunnel soffit was captured by the instrumentation system. This could be a combined consequence of two possible factors. First, the flexural stiffness of the NSL tunnel boxes which might be conservatively estimated during design. Second, the undrained shear strength of the JGP-improved ground which is much higher than the design-required value. As a result, the provision of flat jacks becomes redundant.



(a) (b)

Figure 13. (a) Section Illustrating the Construction Works at Marina Bay Station and (b) 3D view of the ground improvement configuration (from Hashida et al. (2018)).



Figure 14. Excavation face with existing foundation pile during shield driving (from Hashida et al. (2018)).

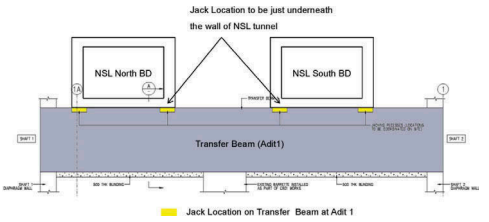


Figure 15. Arrangement of flat jacks between transfer beam and NSL tunnels (from Hashida et al. (2018)).

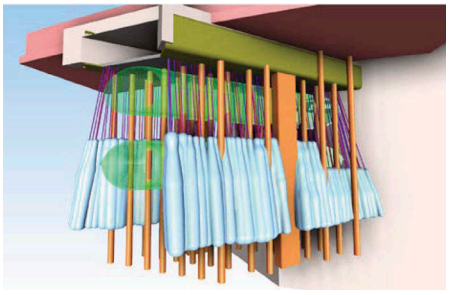


Figure 16. 3D model of the frozen soil based on temperature monitoring data (from Chua et al. (2018)).

Mining of the 5.8m internal diameter stacked mainline tunnels underneath the constructed pedestrian linkway with excavator and temporary ground support was proposed due to the challenging ground condition and the short tunnel span of 40m. This also leads to the adoption of the sprayed concrete lining (SCL) method, which requires dry condition for construction. The lower mainline tunnel is in mixed soil condition of Marine Clay and Old Alluvium. However, since ground improvement is deemed to be ineffective to treat Old Alluvium, there is a risk of water ingress during mined excavation. As such, ground freezing technique is proposed to prevent water ingress through the Old Alluvium layer during mined excavation. Vertical ground freezing tubes were installed from within the completed pedestrian link to the depth of Old Alluvium layer to prevent excessive water ingress and to ensure stability of the mining faces (Figure 16). The permanent lining of the mainline tunnels is constructed by cast-in-situ concrete lining.

The main concern with ground freezing includes soil expansion (heaving) during soil freezing and settlement during thawing. Special laboratory tests were conducted to investigate the behaviour of the frozen local soil, and to determine its engineering properties as well as design parameters. 3D finite element modelling was then carried out to simulate the mining of the

Table 2. Comparison between predicted and actual movement of existing tunnels.

Location of interface	Predicted movement (mm)	Actual movement (mm)
East of Outram Park (Parallel to EWL)	58 *(cumulative displacement by several construction activities)*	2
East of Outram Park (Overcrossing NEL)	4.3	< 2
Undercrossing EWL at Robinson road	55 *(with mitigation measures)*	< 12

mainline tunnels in conjunction with ground freezing. The impact of the construction on the existing MRT tunnels was then investigated and controlled.

The finite element analysis predicted a maximum heave of NSL tunnels to be 7mm and the overall settlement to be 7 mm, considering all construction phases. The vertical displacement of the NSL tunnels was continuously recorded during the construction of the mined tunnels using the ATMS. The actual deformation of the NSL tunnels was significantly smaller than the predicted 7 mm. This is attributed to the fact that the structural rigidity and restraints of the NSL tunnels were not fully incorporated in the finite element model.

In addition, the heaving of the frozen soil was not observed according to the monitoring data. This suggests that the test results of the frozen soil sample may not represent the actual behaviour of the frozen in-situ soil mass.

6 CONCLUSION

The TEL project in its entirety can make a good argument to be one of the most challenging MRT design and construction projects in Singapore and South-East Asia. The alignment encounters all of Singapore's major geological formation and drives into the heart of its highly developed CBD. TSL-D embodies the most complex of engineering challenges, with contrasting geological conditions and close proximity tunnelling to existing railway and stations.

Table 2 summarises the predicted movement of the existing tunnels at interfaces with TSL tunnel alignment and compared against the actual data recorded via the monitoring system. From the table, it can be seen that the design analysis is conservative as the achieved movement with closed face TBM tunnelling control measures and additional considered mitigation measures implemented is much lower.

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